

Sales Analytics & Forecasting Platform

End-to-End Data Engineering, Analytics, Forecasting, and BI

Olist Brazilian E-Commerce Dataset

Documentation Date: May 20, 2026

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1. Introduction

1.1 Brief Introduction

This document describes Project 3: an end-to-end sales analytics and forecasting platform built on the Olist Brazilian E-Commerce public dataset. The marketplace dataset captures real orders, customers, products, payments, and delivery behaviour across Brazil, making it a practical sandbox for data engineering, business intelligence, and machine learning.

The project is implemented as a sequence of Jupyter notebooks and PostgreSQL objects, moving from raw CSV ingestion to a governed fact table, advanced analytics, predictive models, operational monitoring, and Power BI-ready exports. Stakeholders ranging from data engineers to commercial leaders can use the outputs to understand what happened in the past, what is likely to happen next, and where unusual events require intervention.

1.2 Project Objective

The primary objective is to design and demonstrate a production-style analytics workflow that transforms multi-source e-commerce data into actionable business intelligence. Specifically, the project aims to:

- Ingest and model Olist sales data in PostgreSQL with a reusable star-style fact table (fact_sales).
- Deliver SQL and Python analytics covering growth, segmentation, seasonality, and profitability proxies.
- Forecast daily revenue using classical and deep learning time-series methods with rigorous evaluation.
- Detect revenue anomalies and translate them into clear operational recommendations.
- Prepare curated datasets and hierarchies for a multi-page Power BI executive dashboard.
- Document methodology, insights, and future enhancements for handoff to business and IT teams.

1.3 Scope of the Project

The following items are in scope for this engagement:

- Data sources: Nine Olist CSV files (orders, order items, customers, products, payments, reviews, sellers, geolocation, category translations).
- Technology: Python (pandas, SQLAlchemy), PostgreSQL (sales_db), statsmodels/Prophet/TensorFlow for forecasting, Power BI for presentation.
- Time grain: Order and line-item level in the warehouse; daily and monthly aggregates for trends and forecasting.
- Geography: Brazil-wide analysis with state-level (region) and city-level (customer_location) breakdowns where available.
- Analytics domains: Revenue, orders, categories, regions, payments, delivery time, cohorts, RFM, seasonality, forecasts, and anomalies.
- Deliverables: Five-phase notebooks, CSV exports under exports/ and outputs/, and this project documentation.

1.4 Project at a Glance

At completion, the solution processes approximately 112,650 order line items, representing 98,666 distinct orders, and \$13.59 million in total revenue. The dominant state by revenue is São Paulo (SP); leading categories include health & beauty and watches & gifts. Five integrated phases—data modeling, advanced analytics, forecasting, anomaly detection, and BI preparation—form a coherent path from raw files to executive-ready dashboards.

2. Solution Architecture

Architecture Overview

[Figure Placeholder: Data flows from nine Olist CSV sources through PostgreSQL staging tables into fact_sales, then through Python analytics notebooks and exported CSV files consumed by Power BI.]

Figure: Data flows from nine Olist CSV sources through PostgreSQL staging tables into fact_sales, then through Python analytics notebooks and exported CSV files consumed by Power BI.

- Source layer: Nine CSV files (orders, items, customers, products, payments, reviews, sellers, geolocation, category translations).
- Staging layer: Normalized PostgreSQL tables mirroring source entities.
- Mart layer: fact_sales as the primary analytical fact table.
- Analytics layer: Jupyter notebooks (Phases 2–5) for SQL, ML, and exports.
- Presentation layer: Power BI dashboard (four pages) fed by exports/ directory.

3. Phase 1 — Data Loading & SQL Data Modeling

3.1 Objectives

Phase 1 establishes the foundational data pipeline. Raw Olist CSV files are ingested, cleansed, and loaded into PostgreSQL staging tables. A conformed sales fact table (fact_sales) is built to support downstream SQL, Python analytics, and business intelligence.

3.2 Data Ingestion

- Read nine CSV sources using pandas with consistent dtype handling and encoding.
- Map product categories to English labels via product_category_name_translation.
- Derive customer region from state codes and customer_location from city names.
- Construct deliveries, payments, and order line items as relational entities.

3.3 PostgreSQL Staging & Fact Modeling

- Create staging tables: staging_orders, staging_customers, staging_products, staging_payments, and related entities.

- Define fact_sales with composite primary key (order_id, order_item_id).
- Load measures: price, quantity, total_revenue, freight_value, delivery_time, review_score.
- Load dimensions: order_date, customer_location, product_category, payment_type, seller_region.

3.4 Data Quality & Calculated Fields

- Missing values: COALESCE for unknown categories, locations, and payment types; interpolation for sparse daily revenue in later phases.
- Calculated fields: total_revenue = price × quantity; delivery_time from order timestamps.
- Geospatial attributes: customer_lat, customer_lng where available for mapping.
- Assumed margin (30%) used in Phase 2 profit approximations where applicable.

3.5 Core SQL Analyses (Phase 1)

- Daily sales aggregation by order_date.
- Monthly revenue trends with year-over-year comparisons.
- Revenue by product_category and customer region.
- Payment mix and average delivery time by category.
- Top customers and sellers by revenue contribution.

4. Phase 2 — Advanced Analytics & Visualization

4.1 Objectives

Phase 2 extends the warehouse with twelve or more advanced SQL queries, eight or more Python visualizations, seasonality analysis, and narrative business insights. Analytics join fact_sales to staging customers for region-aware metrics.

4.2 Advanced SQL Analytics

- Year-over-year and month-over-month revenue growth (q01, q02).
- Pareto analysis of category revenue concentration (q03).
- Cohort retention by first-order month (q04).
- RFM segmentation: Recency, Frequency, Monetary scoring (q05).
- Category × month heatmaps for seasonality (q06–q07).
- Regional revenue and estimated profit (q08).
- Delivery time distributions by category (q09).
- Payment method trends and order frequency (q10–q12).
- Seasonality blocks: monthly index, quarterly trends, day-of-week patterns, delivery seasonality (q13–q16).

4.3 Visualizations

- Daily and monthly revenue trend lines.
- Category revenue bar charts and Pareto curves.
- Regional revenue and profit comparison charts.

- RFM segment distribution and cohort retention heatmaps.
- Seasonality panels: monthly, quarterly, DOW, delivery-by-month, category $\times$ month heatmap.

4.4 Key Analytical Insights (Phase 2)

- Revenue growth varies materially by year; recent periods require careful interpretation of partial months.
- A small set of categories drives a disproportionate share of total revenue (Pareto effect).
- Cohort retention declines after early months, indicating opportunity for reactivation campaigns.
- RFM segments highlight champions versus at-risk customers for targeted marketing.
- Clear calendar seasonality appears in monthly and day-of-week patterns.

5. Phase 3 — Time-Series Forecasting

5.1 Objectives

Phase 3 models daily revenue as a univariate time series. Classical (ARIMA, Prophet) and deep learning (LSTM) approaches are trained, evaluated, and compared on a held-out test period.

5.2 Time-Series Concepts

- Trend: Long-run level shifts in daily revenue visualized via rolling means.
- Seasonality: Weekly and yearly cycles captured especially by Prophet.
- Stationarity: Augmented Dickey-Fuller (ADF) tests; differencing applied when unit root is present.
- Train-test split: Final 90 days reserved for out-of-sample evaluation.

5.3 Models

- ARIMA: Order selected via `pmdarima auto_arima` when available; forecasts short horizons efficiently.
- Prophet: Handles seasonality and provides uncertainty intervals; component plots for trend/weekly/yearly effects.
- LSTM: Min-max scaled sliding windows (30-day lookahead); simple single-layer architecture as a deep-learning baseline.
- Rolling forecast: Walk-forward one-step ARIMA simulation on the test window.

5.4 Evaluation & Interpretation

- Metrics: MAE (mean absolute error), RMSE (penalizes large errors), MAPE (percentage error).
- Best model selected by lowest RMSE on the test set for the 90-day forward outlook.
- Forecasts exported to `outputs/phase3_test_predictions.csv` for Phase 4 and Phase 5.
- Interpretation: Use confidence bands for planning ranges; point forecasts are planning aids, not guarantees.

6. Phase 4 — Anomaly Detection & Business Recommendations

6.1 Objectives

Phase 4 operationalizes monitoring by flagging unusual daily revenue, comparing detection methods, generating rule-based recommendations, reconciling forecast models, and documenting pipeline and scenario thinking.

6.2 Anomaly Detection Methods

- Z-score method: Flag days where $|z| > 2.5$; classify spikes ($z > 0$) versus drops ($z < 0$).
- Rolling deviation: 30-day rolling mean and standard deviation; flag when $|\text{revenue} - \text{mean}| > 2\sigma$.
- Summary table: date, revenue, anomaly type, detection method exported to phase4_anomaly_summary.csv.

6.3 Business Recommendation Engine

- Enrich anomalies with top product category and region from fact_sales on anomaly dates.
- Rule-based text recommendations for spikes (inventory, staffing) and drops (marketing, ops review).
- Delivery delay correlation: append logistics recommendations when drops coincide with high delivery times.
- Outputs saved to phase4_recommendations.csv.

6.4 Forecast Comparison & Pipeline

- Reload or recompute Phase 3 predictions; MAE, RMSE, MAPE comparison table.
- Overlay plots of actual versus ARIMA, Prophet, and LSTM on the test period.
- Daily pipeline concept: incremental staging load → refresh fact_sales → anomaly job → forecast job → BI refresh.
- Scenario analysis: weekend vs weekday, month-end salary effects, seasonal months, top spike days.

7. Phase 5 — Power BI Dashboard Preparation

7.1 Objectives

Phase 5 produces clean, denormalized CSV exports for a four-page Power BI dashboard without requiring direct database connectivity in the reporting tool.

7.2 Datasets & Exports

- Executive overview: executive_overview.csv (revenue, orders, AOV, MoM %, YoY %).
- Sales trends: daily_sales.csv, monthly_sales.csv, forecast_vs_actual.csv.
- Product insights: top_categories.csv, low_performing_products.csv.
- Regional analysis: sales_by_region.csv, high_growth_regions.csv, sales_by_region_city.csv.

- Modeling aids: date_dimension.csv, region_slicer.csv, category_slicer.csv, date_slicer.csv, hierarchies.csv.

7.3 Power BI Dashboard Pages

- Page 1 — Executive Overview: KPI cards, monthly trend, top region/category snapshots.
- Page 2 — Sales Trends: daily line, monthly columns, forecast vs actual multi-line chart.
- Page 3 — Product Insights: category bars, share donut/treemap, low-performer table.
- Page 4 — Regional Analysis: bar chart,
- Slicers: date range, region, category (synced across pages).
- Hierarchies: Date (Year→Quarter→Month→Day), Product (Category→Product), Region (State→City).

8. Business Insights

The following insights synthesize findings across all phases. They are written for business stakeholders and should be validated against the latest data refresh before executive decisions.

1. Total platform revenue exceeds R\$ 13.5 million across roughly 99,000 orders, with an average order value near R\$ 138—useful for basket-size and pricing benchmarks.
2. São Paulo (SP) accounts for the largest regional revenue share, making it the primary market for campaigns, inventory positioning, and seller recruitment.
3. Rio de Janeiro (RJ) and Minas Gerais (MG) form a strong secondary tier; smaller states represent growth opportunities with tailored logistics and marketing.
4. Health & beauty is the top category by revenue, followed by watches & gifts and bed/bath/table—merchandising should protect stock depth in these leaders.
5. Long-tail categories contribute collectively but individually underperform; assortment reviews can trim low-velocity SKUs.
6. Revenue exhibits calendar seasonality (monthly and day-of-week); promotional calendars should align with historically strong periods identified in Phase 2.
7. Cohort analysis shows retention decay after the first months post-acquisition—lifecycle email and loyalty programs should target month 2–4 drop-off.
8. RFM segmentation distinguishes high-value repeat buyers from one-time purchasers; the budget should shift toward retaining champions and reactivating at-risk segments.
9. Pareto dynamics mean a minority of categories drive a majority of revenue; the supply chain should prioritise inbound flow for the top decile categories.
10. Delivery time varies by category; categories with slow fulfilment correlate with satisfaction risk—carrier SLAs and warehouse slotting need category-aware targets.
11. Payment mix analysis (Phase 2) informs checkout optimisation and instalment strategies for high-ticket categories.

12. Year-over-year revenue growth was positive in the latest comparable year in exported KPIs, but month-over-month figures can be distorted by incomplete current months—report using closed months only.
13. Forecast models (ARIMA, Prophet, LSTM) support 90-day planning horizons; Prophet often excels when weekly seasonality is pronounced, while ARIMA suits smoother series.
14. Z-score and rolling anomaly detectors together reduce false positives: global outliers versus recent baseline breaks require different operational responses.
15. Revenue spikes warrant temporary stock buffers and customer-service surge staffing; sustained spikes may indicate successful campaigns worth repeating.
16. Revenue drops paired with delivery delays suggest logistics root causes before increasing marketing spend.
17. Weekend versus weekday patterns (Phase 4 scenarios) should inform ad scheduling and warehouse shift planning.
18. Month-end revenue lifts may reflect salary-cycle purchasing—time limited promotions accordingly.
19. Power BI exports enable self-service exploration without SQL skills, accelerating regional and category reviews for category managers.
20. Closing the loop—daily incremental loads, anomaly alerts, and forecast refresh—creates a proactive operations rhythm rather than reactive monthly reporting.

9. Future Scope & Enhancements

9.1 Advanced Analytics & Machine Learning

- Gradient boosting (XGBoost, LightGBM) for tabular demand forecasting with exogenous features (promotions, holidays).
- Transformer-based time-series models (Temporal Fusion Transformer) for multi-series category forecasts.
- Customer segmentation via clustering and propensity models for churn and next-best-offer.
- Dynamic pricing experiments informed by elasticity models and competitor signals.

9.2 Productisation & APIs

- REST forecasting API (FastAPI) serving batch and on-demand predictions from retrained models.
- Containerised deployment (Docker/Kubernetes) with model registry (MLflow).
- Embedded analytics in the seller portal for regional partners.
- Mobile-friendly Power BI or custom React dashboards for field teams.

10. Conclusion

This five-phase project demonstrates a mature analytics lifecycle: trustworthy data ingestion and modeling (Phase 1), deep exploratory and segmentation analytics (Phase 2), predictive forecasting (Phase 3), exception-based monitoring and recommendations (Phase 4), and executive-ready BI assets (Phase 5). The architecture is modular—each phase can be refreshed independently as new Olist or production order data arrives.

For production deployment, prioritise automated incremental loads, closed-month KPI definitions, and governed metric catalogues so that business insights remain consistent between SQL, notebooks, and Power BI. The foundation built here supports revenue operations, supply chain planning, and marketing optimization for e-commerce at scale.

Appendix A — Project Artifacts

- `sales_analytics_pipeline.ipynb` — Phase 1
- `sales_analytics_phase2.ipynb` — Phase 2
- `sales_analytics_phase3.ipynb` — Phase 3
- `sales_analytics_phase4.ipynb` — Phase 4
- `sales_analytics_phase5.ipynb` — Phase 5
- `exports/` — Power BI CSV datasets in GitHub
- `outputs/` — Forecast and anomaly intermediate files in GitHub